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ENERGY TRANSFER IN A STRING-COUPLED TWO-PENDULUM SYSTEM: RELATION WITH STRING SLACK AND PENDULUM SEPARATION

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ABSTRACT

Coupled pendulum systems are classic models for studying energy transfer in oscillatory systems. This experiment investigated the beat time (the period of complete energy exchange) between two pendulums coupled by a horizontal cotton string of width 10^{-3}m suspended between vertical rods. Two parameters were systematically varied: string slack (S) (0–0.40 m) and pendulum separation distance (a) (0.04–0.40 m). Beat time was measured using a digital stopwatch, with a standard deviation of 0.367 s. The data follows an exponential relationship $t(S, a) = A \times e^{k}$ $where\ k = \alpha S + \beta a + \gamma Sa$, with fitted coefficients $A \approx 5.89s, \alpha \approx -3.65m^{-1}$, $\beta \approx 5.64m^{-1}$, $\gamma \approx -2.52m^{-2}$. While increasing separation increases beat time as expected, increasing slack decreases beat time—contrary to the initial hypothesis—indicating a transition from tension-by-inertia-mediated to wave-mediated coupling. This interpretation is supported by observations of increased string vibrations at high slack. Anomalous behavior at (S = 0.02, 0.03 m) with large separation reveals a two-cycle energy transfer pattern, suggesting resonant conditions between pendulum frequency and string modes.

INTRODUCTION

Coupled oscillator systems, particularly those including pendulums, serve as models in classical mechanics for exploring complex phenomena, including **non-linearity** and the transition to chaos. Such systems are typically constructed by connecting two or more masses in a manner that permits energy transfer between them, often facilitated by a spring[5]. In a coupled pendulum system, an initial displacement of one pendulum results in a gradual decrease of its amplitude, followed by a growth in the amplitude of the others. This energy is then transferred back to the first oscillator, establishing a recurring cycle. This process appears as a **beat pattern**, a fundamental phenomenon of periodic energy exchange between the coupled oscillators. The resulting complex motion of the system can be deconstructed into two fundamental vibrational patterns, known as **normal modes**: one where the pendulums oscillate in the same direction and another one in the opposite direction[1].

While the theoretical behavior of coupled oscillators is well-established for ideal conditions, assuming perfect energy conservation, and negligible damping, experimental observations often deviate from these ideal predictions. Factors such as air resistance, friction at suspension points, and the non-ideal nature of coupling elements introduce complexities that are not adequately captured by simplified theoretical models[3][5].

The particular system chosen in this study has many variables but two variables that seemed to affect the beat time the most were chosen: Slack of the coupling string and the distance between pendula. Hrishikesh-dada advised us to control variables such as string material and knots which can play an important role.

The present investigation addresses this gap by experimentally characterizing how geometric parameters affect energy transfer in a string-coupled pendulum system. Specifically, this study examines the relationship between coupling strength, determined by the configuration of the connecting element, and the resulting beat period. It is hypothesized that:

1. As coupling slack increases, the coupling strength between pendulums will decrease, resulting in longer beat times between oscillators.

2. As the horizontal separation between pendulums increases, the coupling forces will diminish, also leading to increased beat times and slower energy exchange.

LITERATURE REVIEW

The literature acknowledges nonlinear behavior. Sado (1993) provided a comprehensive survey of energy transfer in nonlinear vibrating systems, noting that not all types of nonlinearities had been widely examined . Particularly relevant is the observation that systems combining polynomial nonlinearities with pendulum geometry (due to the pendulum's presence) have significant applications but remain underexplored[3].

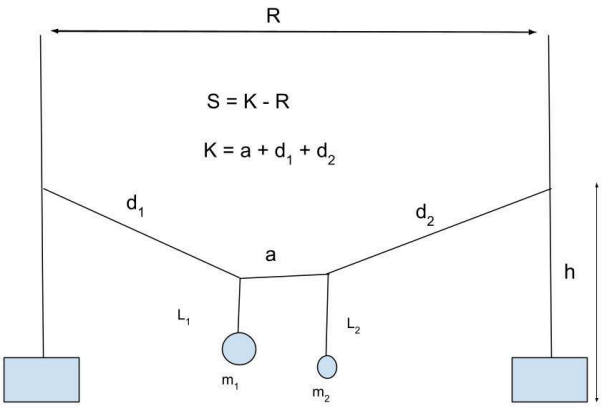
Cziprok, Miron, and Popescu (2015) developed an innovative experimental approach using a thin rod as the coupling element between two pendulums . Their configuration bears important similarities to the present investigation's dual-rod design, with the key distinction that their coupling element is a rigid rod rather than a flexible string. Notably, their experiments investigated how the time of energy transfer between the two bobs varies as a function of the height of the coupling rod.

Sepulchre (2003) reviewed concepts of energy localization in coupled oscillators, focusing on energy barriers in coupled pendula . This theoretical framework addresses how nonlinearity and discreteness can lead to energy localization phenomena, complementing the energy transfer focus of the present investigation[4].

A recent study by Pang et al. (2025) utilizes two pendulums coupled through an elastic band, creating a weakly coupled system suitable for observing both normal modes and beat phenomena. This coupling mechanism allows for controlled energy transfer between the oscillators while maintaining the distinct identities of the individual pendulums. The weak coupling is particularly advantageous for beat observation, as it produces energy exchange cycles without introducing excessive dynamical complexity[2].

METHODOLOGY

Figure 1: Setup Model



Apparatus

Support Structure: Two vertical metal rods secured parallel to each other, mounted perpendicular to the ground at a fixed separation of 5×10^{-1}m.

Coupling String: A lightweight inextensible cotton string of width 10^{-3}m tied horizontally between the two vertical rods.

Pendulums: Two identical pendulums suspended from the horizontal coupling string. Each pendulum consists of a pendulum string of length 0.2m attached to the coupling string. A bob of mass roughly 6.9×10^{-2} kg attached at the free end.

Measurement Tools: Digital Stopwatch; Smartphone Timer.

Variable Analysis

Independent Variables

Pendulum Separation Distance (a):

Distance between both the pendulum strings at the suspension points. It varied from 4×10^{-2}m to 4×10^{-1}m in steps of 2×10^{-2}m.

String Slack (S):

Difference between the length of the horizontal coupling string (K) and the separation distance between the two rods (R). Slack varied from 0m to 2×10^{-1}m in steps of 10^{-2}m.

Dependent Variables

Beat Time (t):

The time required for complete energy transfer from one pendulum to the other and back. Measurements were recorded in seconds and rounded to one decimal place of precision.

Controlled Variables

Pendulum Bob Mass (m_1, m_2):

Both pendulums used bobs of approximately equal mass, measured at 6.9×10^{-2} kg each.

Pendulum Bob Diameter:

Both bobs had similar diameters of approximately .

Pendulum Offset (d_1, d_2):

Distance between the suspension point of a pendulum to the nearest support rod. These distances were maintained symmetrical for both pendulums (d_1=d_2). It changed due to variation in a. Although offset distance changed when the pendulum separation distance was varied, it was not independently controlled but determined by the relationship $d = (R - a)\times 2^{-1}$

Rod Separation Distance (R):

Distance between both the support rods. Fixed at 5×10^{-1}m.

Coupling String Length (K):

Length of the coupling string attached to both the support rods. Not independent but determined by S = K - R.

Pendulum String Length (L_1, L_2):

The length of the strings suspending each bob from the coupling string. Both pendulums used identical string lengths of 2×10^{-1} m, measured from the attachment point to the center of the bob.

Suspension Height (h):

The distance between the coupling string and the ground. Fixed at 5×10^{-1}m.

Pendulum Displacement Angle (i):

The angle at which a pendulum was suspended relative to the coupling string. Set at $\pi\times 6^{-1}$ radians (30°) for all experiments.

Experiment Procedure

The following procedure was repeated for each combination of pendulum separation distance (a) and string slack (S):

Step 1: Configuration

The fixed rod separation distance (R) = 5×10^{-1}m was verified by measuring the horizontal distance between the inner faces of both rods at three different heights—near the top, middle, and bottom, using a metre ruler.

The coupling string was inspected for any signs of fraying, stretching, or damage. A fresh length of lightweight inextensible string was used for each session to ensure consistent material properties.Both pendulums were examined. The pendulum strings were checked for knots, kinks, or stretching.

The suspension height 5×10^{-1}m was verified by measuring the vertical distance from the ground to the coupling string at both rods using a metre ruler. Adjustments were made by raising or lowering the attachment points on the rods until h was consistent.

Step 2: Initialisation

The pendulum separation distance (a) was set by moving the attachment points along the coupling string to the appropriate positions, measured using another string. The string slack

(S) was set by adjusting the length of the coupling string (K) according to the relationship (K = R + S). Once configured, the pendulum offset distances were verified to be symmetrical using the formula, $d = (R - a)\times 2^{-1}$. Both pendulums were brought to rest at their equilibrium positions. The experimenter positioned themselves at eye level with the pendulums to minimise error during observation.

Step 3: Trigger

One of the pendulums was displaced from $\pi\times 6^{-1}$ radians (30°) relative to coupling string by a protractor while the other was held stationary at its equilibrium. The pendulum was released gently to avoid any transverse motion.

Step 4: Time Measurement

The beat time (t) was measured using a digital stopwatch. Timing began at the moment Pendulum 1 was released. The observer watched the amplitude of Pendulum 2, noting its gradual increase and subsequent decrease. The stopwatch was stopped when the amplitude of Pendulum 2 was negligible after one complete energy transfer cycle, i.e after energy had transferred completely to Pendulum 2 and then back to Pendulum 1.

Step 5: Error Elimination

For each configuration of a and S, the measurement was repeated 3 times. Between trials, the pendulums were allowed to come to a complete rest to ensure identical initial conditions. If any trial was disrupted by external factors (e.g., accidental disturbance), the measurement was discarded and repeated.

Step 6: Data Recording

All measurements were recorded immediately in a data table. For each trial, the measured time was rounded to one decimal place of precision (0.1 s). The configuration parameters (a, S, K) were recorded alongside each set of measurements.

DATA ANALYSIS

Raw Data Format

S = 0.07					
a	Reading 1	Reading 2	Reading 3	Average	Sum of Squares of Differences
0.04	5.8	5.1	5.6	5.5	0.26
0.06	6	6	6.2	6.1	0.02666666667
0.08	6.4	6.4	6.4	6.4	0
0.1	7	7	7.3	7.1	0.06
0.12	7.6	7.5	8	7.7	0.14
0.14	8.2	7.6	8.4	8.1	0.3466666667
0.16	9	8.7	9.2	9	0.1266666667
0.18	9.6	9.5	9.8	9.6	0.04666666667
0.2	10.6	10.3	10.8	10.6	0.1266666667
0.22	12.2	11.6	12.6	12.1	0.5066666667

0.24	13.1	13.2	13.5	13.3	0.08666666667
0.26	14.7	14.6	14.9	14.7	0.04666666667
0.28	16.9	16.8	16.9	16.9	0.00666666666
0.3	19.5	19	19.5	19.3	0.1666666667
0.32	21.3	21.3	21.4	21.3	0.00666666666
0.34	25.1	23.9	25.3	24.8	1.146666667
0.36	29.2	29.1	30.3	29.5	0.8866666667
0.38	39.3	37.9	38.9	38.7	1.04
0.4	47.7	46.9	48.2	47.6	0.86
				Sum:	5.886666667

The root of the sum of squares of standard deviation of each condition was added to the difference of total number of experiments and groups (with same conditions) was used to derive standard deviation of 3.67×10^{-2} seconds.

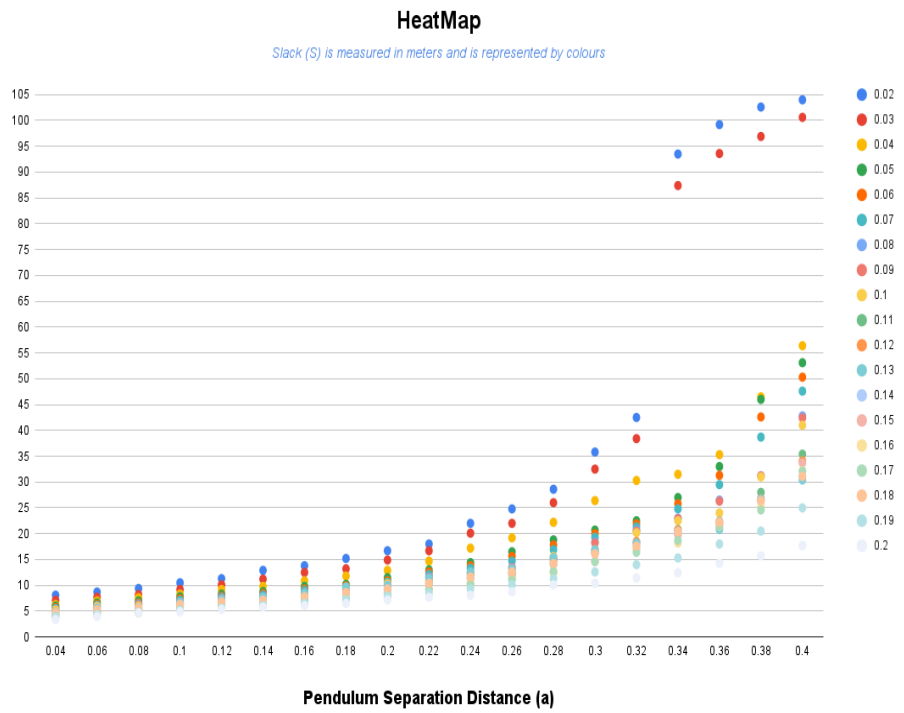
Average Beat Time Data

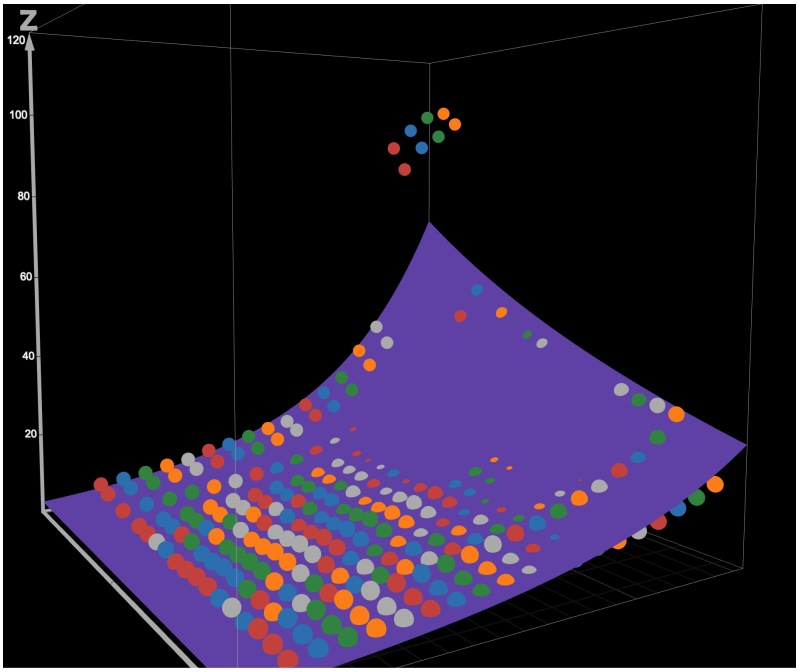
The table given below represents values of average beat time (t) in seconds for different combinations of string slack (S) in meters and pendulum separation distance (a) in meters.

S =0.07		0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.1	0.11	0.12	0.13	0.14	0.15	0.16	0.17	0.18	0.19	0.2
A	0.04	8.1	7.1	6.3	6	5.7	5.5	5.3	5.2	5.2	4.2	4.4	4.8	4.70	4.3	4.5	3.8	4.7	4	3.4
	0.06	8.7	7.7	7	6.6	6.3	6.1	6	5.8	5.9	4.9	5	5.6	5.30	4.9	4.7	4.3	5.3	4.4	4
	0.08	9.4	8.2	7.6	7	6.6	6.4	6.3	6.2	6	5.2	5.3	5.7	5.60	5.2	4.8	4.7	5.6	4.8	4.7
	0.1	10.5	9.2	8.3	7.8	7.4	7.1	6.8	6.7	6.4	5.8	6	6.7	6.30	5.7	5.5	5.5	6.3	5.2	4.9
	0.12	11.3	10.1	9.2	8.3	7.9	7.7	7.5	7.4	7	6.4	6.7	7.3	6.80	6.3	5.7	5.9	6.8	5.6	5.4
	0.14	12.9	11.2	9.8	8.8	8.4	8.1	7.8	7.6	7.5	6.7	6.9	7.7	7.10	6.3	6.4	6.6	7.1	6.1	5.8
	0.16	13.8	12.5	10.9	9.8	9.3	9	8.5	8.5	8.3	7.5	7.8	8.5	7.80	6.6	6.8	7	7.8	6.5	6.1
	0.18	15.2	13.2	11.8	10.2	9.9	9.6	9.5	9.4	9.3	8.3	8.9	9.7	8.60	7.4	7.9	7.5	8.6	7.2	6.5
	0.2	16.7	14.9	12.9	11.5	10.9	10.6	10.2	10	10.3	9	9.3	10	9.30	8	8.6	8.3	9.3	7.9	7.2

	0.22		18	16.7	14.7	13	12.6	12.1	11.6	11.5	11	10.3	10.8	11.4	10.40	9.2	9.2	9.2	10.4	8.5	7.7
	0.24		22	20.1	17.2	14.4	13.8	13.3	12.5	12.3	12.5	11.4	11.9	12.6	11.60	10.3	10.6	10.2	11.6	9.3	8.1
	0.26		24.8	22	19.2	16.5	15.6	14.7	13.5	13.4	13.1	12.2	12.5	13.2	12.60	11.6	11.3	11.1	12.6	10.1	8.8
	0.28		28.6	26	22.2	18.8	17.8	16.9	15.4	15.2	15.6	14.1	14.7	15.4	14.20	12.5	12.4	12.7	14.2	11.3	10.1
	0.3		35.8	32.5	26.4	20.7	20	19.3	18.4	18.2	16.3	16.5	16.9	16.9	16.10	14.7	14.7	14.6	16.1	12.6	10.4
	0.32		42.5	38.4	30.3	22.5	21.9	21.3	20.7	20.5	20.2	18.2	18.5	18.2	17.60	17.1	16.7	16.4	17.6	14	11.4
	0.34		93.5	87.4	31.5	27	25.8	24.8	23	22.9	22.5	20.8	20.7	20.1	20.40	19.8	18.2	18.7	20.4	15.3	12.4
	0.36		99.2	93.6	35.3	33	31.3	29.5	26.5	26.3	24	22.4	21.8	20.8	22.20	22.3	21.6	21.3	22.2	18	14.3
	0.38		102.6	96.9	46.5	46	42.6	38.7	31.2	31.2	31	28	26.8	26.7	26.40	26.2	25.2	24.6	26.4	20.5	15.7
	0.4		104	100.6	56.4	53.1	50.3	47.6	42.8	42.4	41	35.4	34.1	30.4	31.10	33.7	32.2	32.1	31.1	25	17.7

Graphs





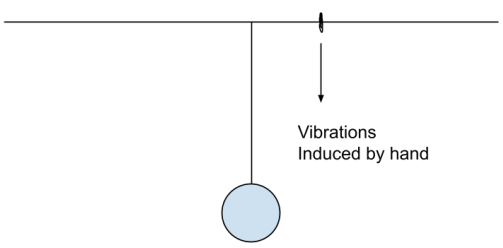
Results

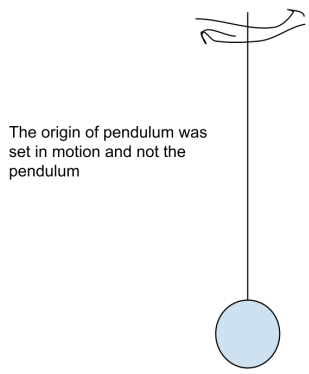
The experimental data reveal systematic relationships between beat time and the two controlled parameters. Effect of Pendulum Separation: Beat time (t) increases monotonically with pendulum separation (a) for all values of string slack (S). The relationship is exponential, with longer beat times usually observed at larger separations with the exception of S = 0.02m and S = 0.03m where consistent large deviations from the overall pattern were observed across multiple trials.Beat time (t) decreases as string slack (S) increases. The decrease follows a near-linear trend for all configurations.

Using computational software (Desmos), the following relationship was derived:

$t(S,a) \approx A \times e^{k}$ where $k = \alpha S + \beta a + \gamma S a$ where
$A \approx 5.89$, $\alpha \approx -3.65$, $\beta \approx 5.64$, $\gamma \approx -2.52$ where all values are in the system of international units.
The fit has a residual error of approximately 5%.

Qualitative Experimenting





The above setups were used to observe individual effects of certain factors such as wave-mediated coupling through the string, inertia during opposing motion in between the pendula or tension in the string changing the equilibria of the pendula. In the first setup, where vibrations were induced on the coupling string, it had an effect which decreased and increased the amplitude of the swinging pendulum, in a similar way to the beat pattern. In the second setup, the stationary pendulum stayed stationary at start due to inertia but slowly accelerated to the speed of the moving origin point. The moving of origin point can be related to the moving of another pendulum, whose mass affects the string tension ultimately changing the equilibrium of the pendulum.

DISCUSSION

Hypothesis Testing

The observed relationships confirm one of the two proposed hypotheses. The increase in beat time(t) with pendulum separation (a) supports the hypothesis that greater distance between pendulums weakens coupling, resulting in slower energy transfer. However, the decrease in beat time with increasing slack (S) contradicts the initial hypothesis that more slack would produce longer beat times. This unexpected result suggests that slack may affect the coupling mechanism in different ways which will be discussed ahead.

Qualitative Experiment Observations

At higher slack values, increased transverse vibrations were observed in the coupling string, particularly near the attachment points to the support rods. These string vibrations indicate the traveling of waves along the coupling element which is shown to have an effect in the qualitative experiments. The moving-origin-stationary-pendulum experiment showed that the movement of the other pendulum moving and affecting the equilibrium doesn't allow for higher energy transfer rate compared to the coupling string vibrations.

Cause-Effect Analysis

Unlike the low-slack regime where the coupling string is more stretched, energy transfers through changes in equilibrium, which forces the pendulum to adjust accordingly. At a high-slack regime, where the coupling string is more relaxed and allows for waves to form. This offers an explanation for the observed decrease in beat time with increasing slack, as wave-mediated coupling can transfer energy more rapidly than the slow beat characteristic of weakly coupled oscillators.

Anomalies at Low Slack, High Pendulum Separation

At low slack values (S=0.02 m and S=0.03 m), particularly when combined with large pendulum separation (a>0.32 m), the measured beat times deviated consistently from the values predicted by the exponential model. In these configurations, the observed beat times were systematically higher than the fitted curve, indicating slower energy transfer than expected.

These deviations were reproducible across multiple trials and appeared consistently for both slack values, suggesting they represent a real physical phenomenon rather than random measurement error. The magnitude of the deviation was approximately 35-45 seconds, corresponding to approximately 70% increase relative to the predicted value.

During the first cycle after release, Pendulum 2 exhibited only a slight motion while Pendulum 1 slowed marginally: far less energy transfer than expected. The second cycle showed the opposite behavior: Pendulum 2 began moving significantly while Pendulum 1 became nearly stationary. Beat time measurements were taken at the completion of this second cycle, when energy had fully transferred from Pendulum 1 to Pendulum 2 and back.

During pilot experiments, this phenomenon was observed at different pendulum separation with the same slack (0.02m) but with different pendulum string lengths. Slack value (0.02m, 0.03m) may be dependent on coupling string properties (coupling, tensile strength, resonance, material) and rod distance. The separation threshold at which this anomaly is observed may be dependent on pendulum properties such as mass and string length. These are the likely causes since the qualitative experiments point towards these.

The effect of moving pendulum mass affecting the string tension causing the equilibrium of the other pendulum to move might also be a factor. Which may cancel out the wave-mediated coupling through phase shift which may slowly align at the end of the first cycle. After the phases are aligned, it allows for very high energy transfer efficiency.

These observations provide evidence for the wave-mediated coupling mechanism proposed in earlier sections. If coupling were purely through tension affecting equilibria of the pendula, there would be no reason for specific slack values to produce anomalous behavior, nor for the separation threshold to depend on pendulum frequency.

This is characteristic of systems with delayed coupling and is not observed in simple spring-coupled oscillators where energy transfer is instantaneous relative to the oscillation period.

The exponential model fitted to the data assumes a smooth, continuous relationship between beat time and the parameters (S) and (a). The presence of these anomalies indicates that the underlying physics is more complex than the model captures. The model works well for the majority of configurations but fails in regions where wave interference effects become significant.

Physical Interpretation of α, β, γ in the equation

$t(S,a) \approx A \times e^{k}$ *where* $k = \alpha S + \beta a + \gamma S a,$

$A \approx 5.89s,\ \alpha \approx -3.56m^{-1},\ \beta \approx 5.64m^{-1}, \gamma \approx -2.52m^{-2}$

Slack Coefficient (α)

The coefficient α controls how beat time responds to changes in string slack (S). $\alpha = -\frac{11}{3}$ s might be a numerical answer. $\alpha = -\frac{cot30^{\circ}}{R}$ may arise from the initial pendulum displacement of 30 degrees.

Separation Coefficient (β)

The coefficient β governs the dependence of beat time on pendulum separation (a). A candidate emerges when expressing this coefficient in terms of the pendulum length. $\beta = \frac{2}{L\sqrt{\pi}}$. Pi appears in Gaussian Integrals used in wave packets (local waves) which might appear in the coupling string.

Interaction Coefficient (γ)

The cross-term coefficient γ captures how slack and separation together influence beat time. Within experimental uncertainty, this value is consistent with: $\gamma = -\frac{1}{4RL}$. This factor is completely geometric and decides how much area over which S and a interact. This likely originates due to trapezoidal geometry of a sagging string.

Prefactor (A)

The prefactor A represents the time scale of the system. $f = \frac{1}{2\pi}\sqrt{\frac{g}{L}}, A = \frac{2\pi}{f}$This might show a relation between the initial displacement angle and natural frequency of the pendula.

CONCLUSION

Summary

This study investigated the energy transfer dynamics in a coupled pendulum system consisting of two pendulums suspended from a horizontal string tied between vertical rods. The beat time (t), the period required for complete energy exchange between pendulums—was measured as a function of two key parameters: string slack (S) and pendulum separation distance (a). Data were collected for (S) ranging from 0 to 0.40 m and (a) from 0.04 to 0.40 m, with a standard deviation 0.39 s.

Findings

The relationship between beat time and system parameters was described by an exponential model.

$$t(S,a) \approx A \times e^{k} \, where \, k = \alpha S + \beta a + \gamma Sa, A \approx 2.59s, \alpha \approx -0.746m^{-1}, \beta \approx 8.113m^{-1}, \gamma \approx -9.455m^{-2}$$

The positive coefficient β confirmed the hypothesis that increasing pendulum separation lengthens beat time. However, the negative coefficient α contradicted the initial hypothesis, revealing that increasing slack decreases beat time— indicating more complex wave-based coupling than simple inertia-based transfer.

Future Work

There were many limitations such as lack of visualisation of string motion during energy transfer, limitations on generalizability due to pendulum masses and lengths. Future Investigations should address these challenges by

increasing measurement density in anomalous region (S = 0.01 m - 0.03 m) and by systematically varying pendulum length and mass to explore constants.

Remarks

The operational definition S = K - R is not the same as the physical sag height. The actual sag. Depends on string length (K) and pendulum separation distance (a). When pendulums are moved farther apart, the string gets pulled more horizontal, reducing the sag even if K stays the same. This means S and a are not truly independent in the real system.

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